ASSIGNMENT - I

A university surveyed 100 students about their course preferences in Math (M), Science (S), and English (E). The following data is collected:

- 55 students like Math. (M = 55)
- 50 students like Science. (S = 50)
- 40 students like English. (E = 40)
- 20 students like both Math and Science. (|M ∩ S| = 20)
- 15 students like both Math and English. (|M ∩ E| = 15)
- 10 students like both Science and English. (|S ∩ E| = 10)
- 5 students like all three subjects. (|M ∩ S ∩ E| = 5)
- 10 students like none of the subjects. (None = 10)

PART - A: Set Theory / Cardinality

1. How many students like only Math?
   - Only Math = |M| - |M ∩ S| - |M ∩ E| + |M ∩ S ∩ E| = 55 - 20 - 15 + 5 = 25

2. How many students like only Science?
   - Only Science = |S| - |M ∩ S| - |S ∩ E|+ |M ∩ S ∩ E| = 50 - 20 - 10 + 5 = 25

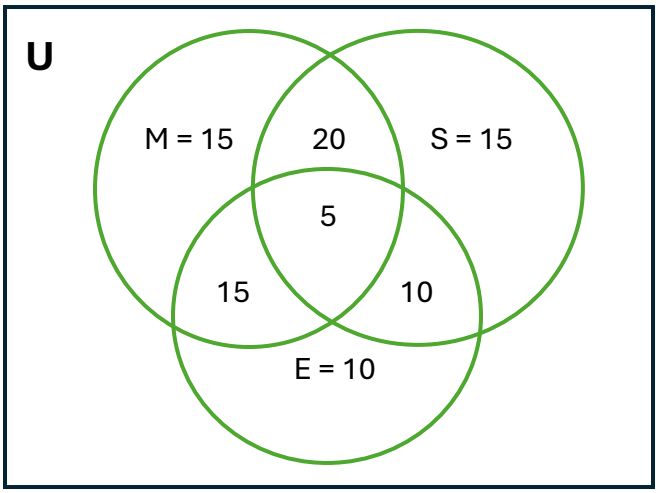
3. How many students like only English?
   - Only English = |E|- |M ∩ E| - |S ∩E| + |M ∩S ∩ E| = 40 - 15 - 10 + 5 = 20

4. How many students like exactly two subjects
   - First we subtract the triple intersection from each pairwise overlap:
     - Math & Science only = |M ∩ S| - |M ∩ S ∩ E| = 20 - 5 = 15
     - Math & English only = |M ∩ E| - |M ∩ S ∩ E| = 15 - 5 = 10
     - Science & English only = |S ∩ E| - |M ∩ S ∩ E| = 10 - 5 = 5
   - Exactly two subjects = 15 + 10 + 5 = 30

5. How many students like at least one subject?
   - At least one = Total - None = 100 - 10 = 90

6. Verify total using Inclusion-Exclusion Principle
   - |M U S U E| = |Only Math| + |Only Science| + |Only English| + |Math and Science only| + |Math and English only| + |Science and English only| + |M ∩ S ∩ E| + |None|
     = 25 + 25 + 20 + 15 + 10 + 5 + 5 + 10 = 115
   - Result not the same as Total students!

PART B: Draw a Venn diagram showing M, S, E with all intersections labeled.



Part C: Write SQL queries as in Part A, and Represent the same data using sets in Python or pandas Data Frame and calculate all cardinalities programmatically.

- First Lets Assume theres an SQL Table students with the following format with the given data

student	Likes_math	Likes_science	Likes_english
Alex	True	False	True
Kaleab	False	False	True
eldana	True	False	False
.....	...	...	...

- Now Assuming this format the sql queries would look something like the following
    - Only math = “Select count(student) from students where likes_math=True and likes_science=False and likes_english=False”
    - Only english = “Select count(student) from students where likes_math=False and likes_science=False and likes_english=True”
    - Only Science = “Select count(student) from students where likes_math=False and likes_science=True and likes_english=False”
    - Exactly Two Subjects = “Select count(student) from students where ((likes_math=True and likes_science=True and likes_english=False) or (likes_math=True and likes_science=False and likes_english=True) or (likes_math=False and likes_science=True and likes_english=True));”
    - At least one subject = “select count(student) from students where (likes_math = True or likes_science = True or likes_english = True);”
    - Total students = “select count(student) from students”

- Using Python

```
total_students = 10
M = 55
S = 50
E = 40
M_and_S = 20
M_and_E = 15
S_and_E = 10
M_and_S_and_E = 5
none = 10

only_M = M - M_and_S - M_and_E + M_and_S_and_E
only_S = S - M_and_S - S_and_E + M_and_S_and_E
only_E = E - M_and_E - S_and_E + M_and_S_and_E
exactly_two = (M_and_S + M_and_E + S_and_E) - 3 * M_and_S_and_E
at_least_one = total_students - none

print("Only Math:", only_M)
print("Only Science:", only_S)
print("Only English:", only_E)
print("Exactly two subjects:", exactly_two)
print("At least one subject:", at_least_one)
print("Check total:", only_M + only_S + only_E + exactly_two + M_and_S_and_E + none)
```



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Assignment

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